

**AI-POWERED TRAIN SAFETY SYSTEM WITH
REAL-TIME CROWD AND OBSTACLE DETECTION**

25-26J-134

Project Proposal Report

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**B.Sc. (Hons) Degree in Information Technology
Specializing in Data Science**

Faculty of Computing

Sri Lanka Institute of Information Technology

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DECLARATION

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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ABSTRACT

The Sri Lankan railway network plays a vital role in daily transportation but faces safety and efficiency challenges such as overcrowded compartments, frequent train accidents, and insufficient real time support for drivers in challenging conditions. This study suggests an AI powered Train Safety and Passenger Guidance System that integrates four main components: Compartment Crowd Density Detection, Platform Compartment Alignment, Object Detection, and Intelligent Driver Decision Support. These modules combined to improve passenger distribution, reduce delays, and improve operational safety across the network.

The Compartment Crowd Density Detection system employs cameras to estimate occupancy levels and guide passengers towards less crowded compartments. Based on this, the Platform Compartment Alignment module integrates real time crowd data from both compartments and platforms to dynamically optimize train stopping positions. Unlike existing systems that rely on fixed stop markers, this component uses AI-driven analysis to align overcrowded compartments with less crowded platform zones, improving boarding efficiency, reducing platform congestion, and minimizing safety risks associated with forceful boarding.

The Object Detection System includes track safety sensors to detect various obstacles under varying visibility conditions. Meanwhile, the Intelligent Driver Decision Support System applies to continuously assess the train's environment and provide timely alerts in critical cases.

According to a preliminary review of related research, indicates that while individual studies address aspects of train safety or passenger flow, few integrate real-time crowd sensing with dynamic train alignment and driver decision support. In consideration of the specific constraints on Sri Lanka's railway system, this study offers a fresh framework for context aware railway safety and passenger guidance.

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
NLP	Natural Language Processing
PCA	Platform Compartment Alignment
YOLOv8	You Only Look Once version 8
KPI	Key Performance Indicator
SUMO	Simulation of Urban MObility
CCTV	Closed Circuit Television

INTRODUCTION

Railway transportation remains a backbone of transportation sector in many countries, including Sri Lanka, but it faces crucial safety, efficiency, and passenger experience challenges. Recent studies increasingly applies artificial intelligence (AI) and data-driven approaches to alleviate risks such as overcrowding, accidents at level crossings, and inefficient driver support system in dynamic conditions. Researches based on safety risk assessment have demonstrated the potential of deep learning and sensor fusion to extend railway safety by enabling accurate obstacle detection and predictive decision making for operators [1], [2].

To effectively develop an AI-powered train safety system, various technical domains must be mastered. These include computer vision and crowd analytics for monitoring crowd distribution on platforms and in compartments, multi-sensor fusion for detecting obstacles under different conditions, and decision support engines for assisting drivers in real time. Reviews of sensing technologies highlight how thermal imaging, using cameras, and wireless tracking can provide actionable insights for crowd and safety management in public transport systems [3].

The state of the art today lies in the growing adoption of AI for railway risk management and automation. For instance, some methods of AI have been successfully applied to railway trespassing detection [4], decision support for locomotive operators [5], and risk assessment at railway stations [6]. However, these efforts typically address unique problems rather than offering an integrated solution.

Previous approaches to platform crowding have relied on infrastructure solutions such as platform edge doors or static train stopping positions, which improve safety but do not dynamically adapt to real time crowd distributions. In the same way, most obstacle detection systems rely on a single sensor modality, limiting their performance in poor visibility. Our proposed system builds upon these studies but introduces a multi-component AI framework that integrates compartment crowd detection, platform compartment alignment, obstacle detection, and driver decision support. In particular, the Platform Compartment Alignment module enhances the state of the art by dynamically aligning train compartments with low-density platform zones using both onboard and platform crowd data; an area where limited prior research exists. This work therefore contributes a novel AI driven passenger flow and safety management strategy tailored to the Sri Lankan railway context.

BACKGROUND AND LITERATURE SURVEY

Artificial Intelligence (AI) has become the key of innovation in railway systems, improving both operational efficiency and passenger safety. Liu et al. [7] proposed an AI-powered scheduling system that combines real time operational data to enhance safety monitoring and optimize train schedules. Their study demonstrated how intelligent algorithms can reduce operational risks and delays, but it did not address compartment-level crowding or platform passenger distribution.

Passenger centered research has explored the role of AI in enhancing service quality. Prosper [8] investigated AI-driven passenger feedback analysis in urban rail systems using natural language processing (NLP) to capture commuter experiences. This work highlighted the importance of integrating passenger perspectives into system design but focused on post journey feedback rather than real time passenger flow optimization.

More directly relevant to crowd management, Sarp et al. [9] introduced the concept of AI-powered digital twin trains, where real time passenger flow data is coupled with predictive modeling to optimize train scheduling and crowd distribution. Their approach demonstrated the value of simulating passenger behavior for congestion management. However, existing methods remain limited in their ability to integrate both onboard compartment density and platform side crowd data.

In summary, while AI has been successfully applied in railway scheduling [7], passenger experience analysis [8], and predictive crowd modeling [9], there is a research gap in real time platform compartment alignment. Current study addresses this gap by integrating compartment level occupancy data with platform crowd density to dynamically optimize stopping positions, thereby improving passenger safety and flow efficiency.

RESEARCH GAP

Despite recent advances in AI-powered railway safety and passenger management, several important gaps remain unaddressed.

- **Lack of Integrated Passenger Flow Modeling**

Existing studies mainly focus on either scheduling optimization [7] or passenger feedback analysis [8]. However, there is limited research that integrates onboard compartment crowd density with platform side passenger distribution to optimize train stopping positions in real time.

- **Static vs. Dynamic Platform Alignment**

Most crowd management systems rely on fixed infrastructure, such as platform edge doors or static stop markers, which fail to adapt to changing passenger flows. Few systems explore dynamic platform compartment alignment, where stopping positions are adjusted continuously based on live data [9].

- **Passenger Centric vs. System Centric Approaches**

Current AI applications often prioritize operational efficiency (such as timetabling, scheduling) over passenger boarding safety and comfort. There is a need for solutions that balance system performance with passenger centered safety in congested railway networks.

- **Limited Contextual Application in Developing Railways**

Most previous research has been conducted in metro systems of developed countries, which benefit from advanced infrastructure and automation. There is a lack of studies tailored to developing country railways, where overcrowding, mixed traffic, and limited safety mechanisms pose unique challenges.

Addressing these challenges, our study proposes an AI powered Platform Compartment Alignment module that combines real time compartment occupancy data with platform crowd density to dynamically optimize train stopping positions. This integration is expected to improve passenger flow efficiency, reduce boarding collisions, and complement existing AI driven safety systems.

RESEARCH PROBLEM

Although significant progress has been made in applying artificial intelligence (AI) and sensor technologies to railway safety and passenger management, several gaps remain.

Integration of Compartment and Platform Data - Existing studies often focus on either compartment crowd estimation or platform crowd management separately. There is little research on combining these two data sources to inform dynamic train alignment decisions in real time.

Dynamic vs. Static Alignment Approaches - Current solutions rely on fixed stop infrastructure that cannot adapt to fluctuating crowd densities. Study on dynamic alignment strategies using live crowd sensing is limited.

Passenger Centric Optimization - Most AI based railway researches emphasize operational efficiency (such as scheduling, timetabling, and automation) rather than passenger safety, comfort, and equitable distribution during boarding, especially in overcrowded networks.

Contextual Application to Developing Railways - Prior research is concentrated on metro systems in developed countries with advanced infrastructure. There is a lack of studies addressing the unique challenges of developing country railways, such as Sri Lanka, where overcrowding, train-animal collisions, and minimal automation demand tailored solutions.

OBJECTIVES

Main Objective

To design and implement an AI powered train safety and passenger guidance system that combines real time crowd density detection, platform compartment alignment, obstacle detection, and driver decision support to extend the ability of railway safety, optimize passenger flow, and improve operational efficiency in the Sri Lankan railway network.

Specific Objectives

- To integrate compartment level crowd density data with platform side passenger distribution.

Combining real time passenger occupancy data from the compartment crowd density detection module with camera based crowd analysis of the platform. By fusing these two datasets, the system will create a real time model of passenger flow, identifying mismatches between overcrowded compartments and congested platform areas.

- To develop a dynamic alignment model for optimal stopping positions.

Implementing an AI driven alignment algorithm which is able to calculate where each compartment should stop along the platform. Unlike traditional static stop markers, this model will adjust stopping positions by matching crowded compartments with platform sections that have lower passenger density. The model will use optimization techniques to minimize boarding congestion, improve flow efficiency, and ensure balanced passenger distribution across the train.

- To design a passenger guidance mechanism for effective communication.

Addressing the usability aspect of the system by developing communication channels to guide passengers towards optimal boarding zones. The mechanism includes mobile applications (or audio announcements) which display real time guidance. So that passengers can adjust their boarding behavior, reducing forceful boarding and improving safety.

- To evaluate the effectiveness of alignment strategies through testing.

Evaluating the effectiveness of alignment strategies through testing

The impact of the system must be validated through either simulation based experiments or small scale real world trials. Key performance indicators (KPIs) such as boarding time, platform congestion levels, compartment load balance, and passenger safety will be measured.

METHODOLOGY

Overall System Architecture

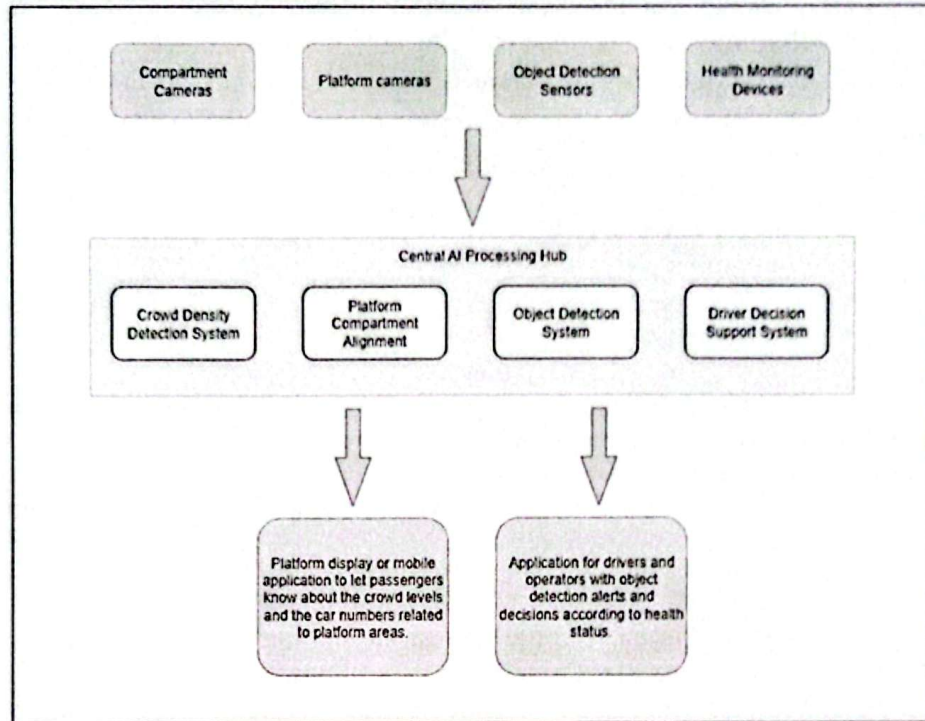


Figure 1: System Architecture Diagram

The proposed system architecture consists of four interconnected AI powered components designed to enhance railway safety and passenger management. The Compartment Crowd Density Detection module uses cameras to monitor passenger occupancy within train compartments in real time. This data directly informs the Platform Compartment Alignment module, which integrates compartment level occupancy with platform side crowd analysis to dynamically determine optimal stopping positions, ensuring efficient and safe boarding. In parallel, the Object Detection System employs sensors to identify potential hazards and transmitting alerts to both the driver and the control center. The Driver Decision Support System integrates inputs from all modules, to calculate contextual risk scores and provide drivers with real time recommendations. In summary, these components form a unified architecture that improves passenger flow, minimizes boarding hazards, prevents collisions, and supports safer decision-making in railway operations.

Platform Compartment Alignment

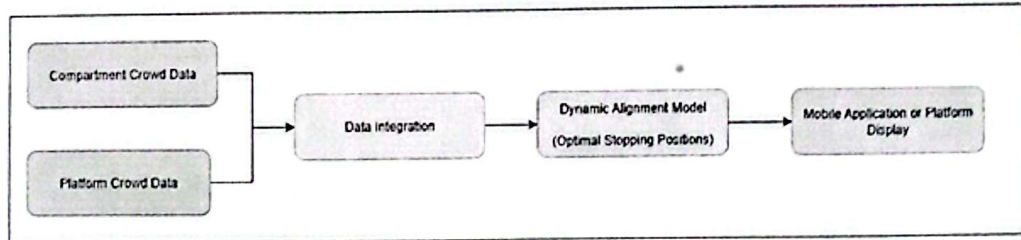


Figure 2: Platform Compartment Alignment Component - High Level System Architecture Diagram

The Platform Compartment Alignment (PCA) module operates as the bridge between the Compartment Crowd Density Detection system and the Passenger Guidance Interface. Real-time compartment occupancy levels (from CCTV/CO₂/weight sensors) are combined with platform-side crowd density data (from CCTV cameras). This integrated dataset is processed by an AI-driven alignment model, which dynamically determines optimal train stopping positions, ensuring that overcrowded compartments align with less congested platform zones. Results are then communicated to passengers via digital displays, platform signage, or mobile notifications, directing them toward safer, less congested boarding areas.

Project Execution Plan

The project will be carried out in several phases:

Task 1: Data Acquisition and Preprocessing

Collect real-time video data from CCTV cameras positioned on platforms.

Obtain compartment occupancy data from the crowd density detection system.

Preprocess video frames using computer vision (YOLOv8) for crowd detection and density estimation.

Task 2: Data Integration

Develop a fusion algorithm to combine onboard occupancy data with platform crowd distribution.

Synchronize datasets with time-stamped identifiers to ensure alignment decisions are made in real time.

Task 3: Dynamic Alignment Model Development

Implement optimization techniques (linear programming, rule-based systems, or reinforcement learning).

Define constraints such as train length, platform length, and passenger safety thresholds.

Simulate alignment outcomes under varying crowd conditions.

Task 4: Passenger Guidance Mechanism

Prototype a mobile application or platform display for stations.

Task 5: Evaluation and Testing

Simulate scenarios with varying crowd densities using datasets or synthetic passenger models.

Measure performance indicators such as boarding time, load balance, and platform congestion reduction.

Validate results against baseline methods .

Resources Required

Equipment: Cameras, Sensors, computers with GPU support for model training.

Software: Python (TensorFlow/PyTorch), optimization libraries, simulation platforms (AnyLogic, SUMO).

Data: Real or recorded CCTV footage from Sri Lankan railway platforms; synthetic datasets where access is restricted.

Support: Access to station layouts, train schedules, and sensor integration infrastructure.

Data Collection

Primary Data: Real-time CCTV footage of platforms and trains.

Secondary Data: Historical passenger volume data from Sri Lanka Railways.

PROJECT REQUIREMENTS

This section outlines the functional and non-functional requirements of the Platform Compartment Alignment component, ensuring that the design aligns with both technical goals and user needs

Functional Requirements

- System must collect real-time compartment crowd data and platform crowd density data.
- Systems must calculate optimal stopping positions dynamically.
- System must provide passenger guidance outputs
- System must log performance metrics and support testing.

Non Functional Requirements

- Performance - Deliver alignment recommendations within few seconds.
- Scalability - Support multiple stations, different train lengths.
- Reliability - Ensure handling missing data.
- Usability - Guidance must be clear, intuitive, accessible.
- Security - Protect data privacy and prevent unauthorized access.
- Maintainability - Allow easy updates and logging for audits.

TIMELINE & SCHEDULE

2025/26 July	Brainstorming workshop
25 Mar	Group registration
26 th May	Topic Assessment Form (TAF)
30 June -16 July May	Project charter
23 July	Proposal reports (draft)
15 Aug	Proposal presentation
8-12 Sep	Proposal reports (final-for marking)
19 September	Progress Presentation - I
15-19 Dec	Check List I
Mar	Research paper
Apr	Final reports
Mar	Progress Presentation - II
Mar	Check List II
June	Submission of RP- Acceptance notification
May	Final presentation & VIVA
June	Project website
June	Research logbook

Figure 3: Timeline

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